

# Homework 8

## APPM 4490/5490 Theory of Machine Learning, Spring 2026

**Due date:** Friday, Mar 13 '27, before 11 AM, via paper or via Gradescope

**Instructor:** Prof. Becker

**Revision date:** 3/6/2026

**Theme:** VC Dimension practice for midterm

**Instructions** Collaboration with your fellow students is OK and in fact recommended, although direct copying is not allowed. The internet is allowed for basic tasks (e.g., looking up definitions on wikipedia) but it is not permissible to search for proofs or to *post* requests for help on forums such as <http://math.stackexchange.com/> or to look at solution manuals; see also the class **AI policy** on our website. Please write down the names of the students that you worked with.

An arbitrary subset of these questions will be graded.

**Reading** You are responsible for reading chapter 10 about “boosting” of **Understanding Machine Learning** by Shai Shalev-Shwartz and Shai Ben-David (2014, Cambridge University Press), as well as chapter 7 in **Foundations of Machine Learning**, 2nd edition, by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar.

**Problem 1:** Show that the VC dimension of decision stumps in  $d$  dimensions is bounded  $\text{VCdim}(\mathcal{H}) \leq c_1 + c_2 \log_2(d)$  for some constants  $c_1$  and  $c_2$ . (Any value of constants work, but you might try to get  $c_1 = 6 - 4 \log_2(\ln(2)) \approx 8.12$  and  $c_2 = 2$ , or better). Note: Exercise 9.5 in Mohri et al. shows more generally that binary Decision Trees of  $k$  nodes in  $d$  dimensions have  $\text{VCdim}(\mathcal{H}) = \Omega(k \log(d))$ .

*Hint:* After picking a dimension, how many dichotomies can a stump achieve? It may help to think of the ERM algorithm we derived for decision stumps; also, decision stumps have nice graphical interpretations, especially for  $d = 2$ . Use this to bound the size of the growth function  $\tau(m)$ , and then do the usual trick: argue that if  $\text{VCdim} \geq m$ , then we must have  $\tau(m) \geq 2^m$ , and use Lemma A.2<sup>1</sup>.

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<sup>1</sup>**Lemma A.2:** If  $a \geq 1, b > 0$  then

$$(x \geq 4a \log(2a) + 2b) \implies (x \geq a \log(x) + b)$$

and (this version isn't in the book)

$$(x > 4a \log(2a) + 2b) \implies (x > a \log(x) + b).$$

These are exactly true for the natural log, but you can use them for  $\log_2$  if needed (where the constants 4 and 2 may to be adjusted but don't worry about that).